

elementary activities a_m , $m = 1, \dots, 8$. With the usual sign convention, the numerical values of these activities are

$$a_1 = (-3, -6, 4, 0)$$

$$a_2 = (-7, -9, 3, 2)$$

$$a_3 = (-1, -2, 3, -1)$$

$$a_4 = (-8, -13, 3, 1)$$

$$a_5 = (-11, -19, 12, 0)$$

$$a_6 = (-4, -3, -2, 5)$$

$$a_7 = (-8, -5, 0, 10)$$

$$a_8 = (-2, -4, 5, 2)$$

It is assumed that any activity can be operated at any nonnegative level $x_m \geq 0$ and that all activities can operate simultaneously at any scale (i.e., for any $x_m \geq 0$, $m = 1, \dots, 8$, the production $\sum_m x_m a_m$ is feasible).

- Define the corresponding production set Y , and show that it is convex.
- Verify the no-free-lunch property.
- Verify that Y does *not* satisfy the free-disposal property. The free-disposal property would be satisfied if we added new elementary activities to our list. How would you choose them (given specific numerical values)?
- Show by direct comparison of a_1 with a_5 , a_2 with a_4 , a_3 with a_6 , and a_6 with a_7 that four of the elementary activities are not efficient.
- Show that a_1 and a_2 are inefficient by exhibiting two positive linear combinations of a_3 and a_7 that dominate a_1 and a_2 , respectively.
- Could you venture a complete description of the set of efficient production vectors?
- Suppose that the amounts of the four goods available as initial resources to the firm are

$$s_1 = 480, \quad s_2 = 300, \quad s_3 = 0, \quad s_4 = 0.$$

Subject to those limitations on the net use of resources, the firm is interested in maximizing the net production of the third good. How would you set up the problem as a linear program?

- By using all the insights you have gained on the set of efficient production vectors, can you solve the optimization problem in (g)? [Hint: It can be done graphically.]

Choice Under Uncertainty

6.A Introduction

In previous chapters, we studied choices that result in perfectly certain outcomes. In reality, however, many important economic decisions involve an element of risk. Although it is formally possible to analyze these situations using the general theory of choice developed in Chapter 1, there is good reason to develop a more specialized theory: Uncertain alternatives have a structure that we can use to restrict the preferences that "rational" individuals may hold. Taking advantage of this structure allows us to derive stronger implications than those based solely on the framework of Chapter 1.

In Section 6.B, we begin our study of choice under uncertainty by considering a setting in which alternatives with uncertain outcomes are describable by means of objectively known probabilities defined on an abstract set of possible outcomes. These representations of risky alternatives are called *lotteries*. In the spirit of Chapter 1, we assume that the decision maker has a rational preference relation over these lotteries. We then proceed to derive the *expected utility theorem*, a result of central importance. This theorem says that under certain conditions, we can represent preferences by an extremely convenient type of utility function, one that possesses what is called the *expected utility form*. The key assumption leading to this result is the *independence axiom*, which we discuss extensively.

In the remaining sections, we focus on the special case in which the outcome of a risky choice is an amount of money (or any other one-dimensional measure of consumption). This case underlies much of finance and portfolio theory, as well as substantial areas of applied economics.

In Section 6.C, we present the concept of *risk aversion* and discuss its measurement. We then study the comparison of risk aversions both across different individuals and across different levels of an individual's wealth.

Section 6.D is concerned with the comparison of alternative distributions of monetary returns. We ask when one distribution of monetary returns can unambiguously be said to be "better" than another, and also when one distribution can be said to be "more risky than" another. These comparisons lead, respectively, to the concepts of *first-order* and *second-order stochastic dominance*.

In Section 6.E, we extend the basic theory by allowing utility to depend on *states of nature* underlying the uncertainty as well as on the monetary payoffs. In the process, we develop a framework for modeling uncertainty in terms of these underlying states. This framework is often of great analytical convenience, and we use it extensively later in this book.

In Section 6.F, we consider briefly the theory of *subjective probability*. The assumption that uncertain prospects are offered to us with known objective probabilities, which we use in Section 6.B to derive the expected utility theorem, is rarely descriptive of reality. The subjective probability framework offers a way of modeling choice under uncertainty in which the probabilities of different risky alternatives are not given to the decision maker in any objective fashion. Yet, as we shall see, the theory of subjective probability offers something of a rescue for our earlier objective probability approach.

For further reading on these topics, see Kreps (1988) and Machina (1987). Diamond and Rothschild (1978) is an excellent sourcebook for original articles.

6.B Expected Utility Theory

We begin this section by developing a formal apparatus for modeling risk. We then apply this framework to the study of preferences over risky alternatives and to establish the important expected utility theorem.

Description of Risky Alternatives

Let us imagine that a decision maker faces a choice among a number of risky alternatives. Each risky alternative may result in one of a number of possible *outcomes*, but which outcome will actually occur is uncertain at the time that he must make his choice.

Formally, we denote the set of all possible outcomes by C .¹ These outcomes could take many forms. They could, for example, be consumption bundles. In this case, $C = X$, the decision maker's consumption set. Alternatively, the outcomes might take the simpler form of monetary payoffs. This case will, in fact, be our leading example later in this chapter. Here, however, we treat C as an abstract set and therefore allow for very general outcomes.

To avoid some technicalities, we assume in this section that the number of possible outcomes in C is finite, and we index these outcomes by $n = 1, \dots, N$.

Throughout this and the next several sections, we assume that the probabilities of the various outcomes arising from any chosen alternative are *objectively known*. For example, the risky alternatives might be monetary gambles on the spin of an unbiased roulette wheel.

The basic building block of the theory is the concept of a *lottery*, a formal device that is used to represent risky alternatives.

Definition 6.B.1: A *simple lottery* L is a list $L = (p_1, \dots, p_N)$ with $p_n \geq 0$ for all n and $\sum_n p_n = 1$, where p_n is interpreted as the probability of outcome n occurring.

1. It is also common, following Savage (1954), to refer to the elements of C as *consequences*.

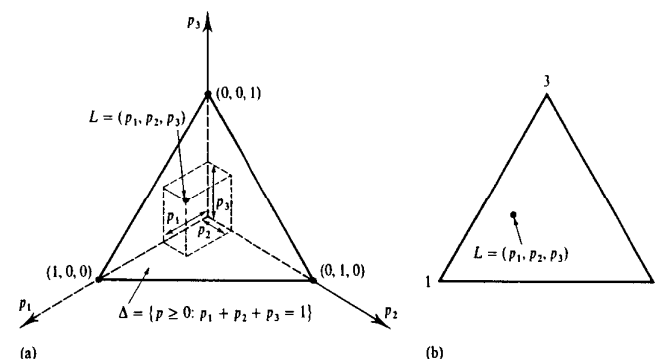


Figure 6.B.1
Representations of the simplex when $N = 3$.
(a) Three-dimensional representation.
(b) Two-dimensional representation.

A simple lottery can be represented geometrically as a point in the $(N - 1)$ dimensional simplex, $\Delta = \{p \in \mathbb{R}_+^N: p_1 + \dots + p_N = 1\}$. Figure 6.B.1(a) depicts this simplex for the case in which $N = 3$. Each vertex of the simplex stands for the degenerate lottery where one outcome is certain and the other two outcomes have probability zero. Each point in the simplex represents a lottery over the three outcomes. When $N = 3$, it is convenient to depict the simplex in two dimensions, as in Figure 6.B.1(b), where it takes the form of an equilateral triangle.²

In a simple lottery, the outcomes that may result are certain. A more general variant of a lottery, known as a *compound lottery*, allows the outcomes of a lottery themselves to be simple lotteries.³

Definition 6.B.2: Given K simple lotteries $L_k = (p_k^1, \dots, p_k^N)$, $k = 1, \dots, K$, and probabilities $\alpha_k \geq 0$ with $\sum_k \alpha_k = 1$, the *compound lottery* $(L_1, \dots, L_K; \alpha_1, \dots, \alpha_K)$ is the risky alternative that yields the simple lottery L_k with probability α_k for $k = 1, \dots, K$.

For any compound lottery $(L_1, \dots, L_K; \alpha_1, \dots, \alpha_K)$, we can calculate a corresponding *reduced lottery* as the simple lottery $L = (p_1, \dots, p_N)$ that generates the same ultimate distribution over outcomes. The value of each p_n is obtained by multiplying the probability that each lottery L_k arises, α_k , by the probability p_k^n that outcome n arises in lottery L_k , and then adding over k . That is, the probability of outcome n in the reduced lottery is

$$p_n = \alpha_1 p_1^n + \dots + \alpha_K p_K^n$$

2. Recall that equilateral triangles have the property that the sum of the perpendiculars from any point to the three sides is equal to the altitude of the triangle. It is therefore common to depict the simplex when $N = 3$ as an equilateral triangle with altitude equal to 1 because by doing so, we have the convenient geometric property that the probability p_n of outcome n in the lottery associated with some point in this simplex is equal to the length of the perpendicular from this point to the side opposite the vertex labeled n .

3. We could also define compound lotteries with more than two stages. We do not do so because we will not need them in this chapter. The principles involved, however, are the same.

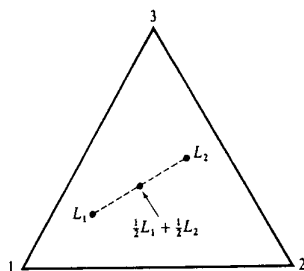


Figure 6.B.2
The reduced lottery of
a compound lottery.

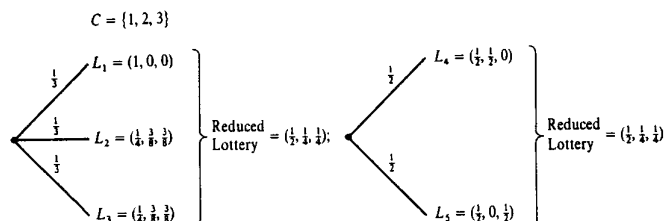


Figure 6.B.3
Two compound
lotteries with the same
reduced lottery.

for $n = 1, \dots, N$.⁴ Therefore, the reduced lottery L of any compound lottery $(L_1, \dots, L_K; \alpha_1, \dots, \alpha_K)$ can be obtained by vector addition:

$$L = \alpha_1 L_1 + \dots + \alpha_K L_K \in \Delta.$$

In Figure 6.B.2, two simple lotteries L_1 and L_2 are depicted in the simplex Δ . Also depicted is the reduced lottery $\frac{1}{2}L_1 + \frac{1}{2}L_2$ for the compound lottery $(L_1, L_2; \frac{1}{2}, \frac{1}{2})$ that yields either L_1 or L_2 with a probability of $\frac{1}{2}$ each. This reduced lottery lies at the midpoint of the line segment connecting L_1 and L_2 . The linear structure of the space of lotteries is central to the theory of choice under uncertainty, and we exploit it extensively in what follows.

Preferences over Lotteries

Having developed a way to model risky alternatives, we now study the decision maker's preferences over them. The theoretical analysis to follow rests on a basic *consequentialist* premise: We assume that for any risky alternative, only the reduced lottery over final outcomes is of relevance to the decision maker. Whether the probabilities of various outcomes arise as a result of a simple lottery or of a more complex compound lottery has no significance. Figure 6.B.3 exhibits two different compound lotteries that yield the same reduced lottery. Our consequentialist hypothesis requires that the decision maker view these two lotteries as equivalent.

4. Note that $\sum_n p_n = \sum_k \alpha_k (\sum_n p_n^k) = \sum_k \alpha_k = 1$.

We now pose the decision maker's choice problem in the general framework developed in Chapter 1 (see Section 1.B). In accordance with our consequentialist premise, we take the set of alternatives, denoted here by $\mathcal{L}$, to be the set of all simple lotteries over the set of outcomes C . We next assume that the decision maker has a rational preference relation $\succeq$ on $\mathcal{L}$, a complete and transitive relation allowing comparison of any pair of simple lotteries. It should be emphasized that, if anything, the rationality assumption is stronger here than in the theory of choice under certainty discussed in Chapter 1. The more complex the alternatives, the heavier the burden carried by the rationality postulates. In fact, their realism in an uncertainty context has been much debated. However, because we want to concentrate on the properties that are specific to uncertainty, we do not question the rationality assumption further here.

We next introduce two additional assumptions about the decision maker's preferences over lotteries. The most important and controversial is the *independence axiom*. The first, however, is a continuity axiom similar to the one discussed in Section 3.C.

Definition 6.B.3: The preference relation $\succeq$ on the space of simple lotteries $\mathcal{L}$ is *continuous* if for any $L, L', L'' \in \mathcal{L}$, the sets

$$\{\alpha \in [0, 1]: \alpha L + (1 - \alpha)L' \succeq L''\} \subset [0, 1]$$

and

$$\{\alpha \in [0, 1]: L'' \succeq \alpha L + (1 - \alpha)L'\} \subset [0, 1]$$

are closed.

In words, continuity means that small changes in probabilities do not change the nature of the ordering between two lotteries. For example, if a "beautiful and uneventful trip by car" is preferred to "staying home," then a mixture of the outcome "beautiful and uneventful trip by car" with a sufficiently small but positive probability of "death by car accident" is still preferred to "staying home." Continuity therefore rules out the case where the decision maker has lexicographic ("safety first") preferences for alternatives with a zero probability of some outcome (in this case, "death by car accident").

As in Chapter 3, the continuity axiom implies the existence of a utility function representing $\succeq$, a function $U: \mathcal{L} \rightarrow \mathbb{R}$ such that $L \succeq L'$ if and only if $U(L) \geq U(L')$. Our second assumption, the independence axiom, will allow us to impose considerably more structure on $U(\cdot)$.⁵

Definition 6.B.4: The preference relation $\succeq$ on the space of simple lotteries $\mathcal{L}$ satisfies the *independence axiom* if for all $L, L', L'' \in \mathcal{L}$ and $\alpha \in (0, 1)$ we have

$$L \succeq L' \text{ if and only if } \alpha L + (1 - \alpha)L'' \succeq \alpha L' + (1 - \alpha)L''.$$

In other words, if we mix each of two lotteries with a third one, then the preference ordering of the two resulting mixtures does not depend on (is *independent* of) the particular third lottery used.

5. The independence axiom was first proposed by von Neumann and Morgenstern (1944) as an incidental result in the theory of games.

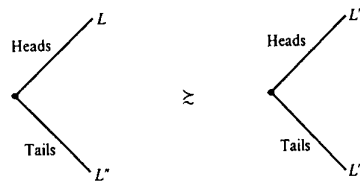


Figure 6.B.4
The independence
axiom.

Suppose, for example, that $L \succsim L'$ and $\alpha = \frac{1}{2}$. Then $\frac{1}{2}L + \frac{1}{2}L''$ can be thought of as the compound lottery arising from a coin toss in which the decision maker gets L if heads comes up and L'' if tails does. Similarly, $\frac{1}{2}L' + \frac{1}{2}L''$ would be the coin toss where heads results in L' and tails results in L'' (see Figure 6.B.4). Note that conditional on heads, lottery $\frac{1}{2}L + \frac{1}{2}L''$ is at least as good as lottery $\frac{1}{2}L' + \frac{1}{2}L''$; but conditional on tails, the two compound lotteries give identical results. The independence axiom requires the sensible conclusion that $\frac{1}{2}L + \frac{1}{2}L''$ be at least as good as $\frac{1}{2}L' + \frac{1}{2}L''$.

The independence axiom is at the heart of the theory of choice under uncertainty. It is unlike anything encountered in the formal theory of preference-based choice discussed in Chapter 1 or its applications in Chapters 3 to 5. This is so precisely because it exploits, in a fundamental manner, the structure of uncertainty present in the model. In the theory of consumer demand, for example, there is no reason to believe that a consumer's preferences over various bundles of goods 1 and 2 should be independent of the quantities of the other goods that he will consume. In the present context, however, it is natural to think that a decision maker's preference between two lotteries, say L and L' , should determine which of the two he prefers to have as part of a compound lottery *regardless* of the other possible outcome of this compound lottery, say L'' . This other outcome L'' should be irrelevant to his choice because, in contrast with the consumer context, he does not consume L or L' together with L'' but, rather, only *instead* of it (if L or L' is the realized outcome).

Exercise 6.B.1: Show that if the preferences $\succsim$ over $\mathcal{L}$ satisfy the independence axiom, then for all $\alpha \in (0, 1)$ and $L, L', L'' \in \mathcal{L}$ we have

$$L \succ L' \text{ if and only if } \alpha L + (1 - \alpha)L'' \succ \alpha L' + (1 - \alpha)L''$$

and

$$L \sim L' \text{ if and only if } \alpha L + (1 - \alpha)L'' \sim \alpha L' + (1 - \alpha)L''.$$

Show also that if $L \succ L'$ and $L'' \succ L'''$, then $\alpha L + (1 - \alpha)L'' \succ \alpha L' + (1 - \alpha)L'''$.

As we will see shortly, the independence axiom is intimately linked to the representability of preferences over lotteries by a utility function that has an *expected utility form*. Before obtaining that result, we define this property and study some of its features.

Definition 6.B.5: The utility function $U: \mathcal{L} \rightarrow \mathbb{R}$ has an *expected utility form* if there is an assignment of numbers $(u_1, \dots, u_N)$ to the N outcomes such that for every simple lottery $L = (p_1, \dots, p_N) \in \mathcal{L}$ we have

$$U(L) = u_1 p_1 + \dots + u_N p_N.$$

A utility function $U: \mathcal{L} \rightarrow \mathbb{R}$ with the expected utility form is called a *von Neumann–Morgenstern (v.N–M) expected utility function*.

Observe that if we let L^n denote the lottery that yields outcome n with probability one, then $U(L^n) = u_n$. Thus, the term *expected utility* is appropriate because with the v.N–M expected utility form, the utility of a lottery can be thought of as the expected value of the utilities u_n of the N outcomes.

The expression $U(L) = \sum_n u_n p_n$ is a general form for a *linear function in the probabilities* $(p_1, \dots, p_N)$. This linearity property suggests a useful way to think about the expected utility form.

Proposition 6.B.1: A utility function $U: \mathcal{L} \rightarrow \mathbb{R}$ has an expected utility form if and only if it is *linear*, that is, if and only if it satisfies the property that

$$U\left(\sum_{k=1}^K \alpha_k L_k\right) = \sum_{k=1}^K \alpha_k U(L_k) \quad (6.B.1)$$

for any K lotteries $L_k \in \mathcal{L}$, $k = 1, \dots, K$, and probabilities $(\alpha_1, \dots, \alpha_K) \geq 0$, $\sum_k \alpha_k = 1$.

Proof: Suppose that $U(\cdot)$ satisfies property (6.B.1). We can write any $L = (p_1, \dots, p_N)$ as a convex combination of the degenerate lotteries $(L^1, \dots, L^N)$, that is, $L = \sum_n p_n L^n$. We have then $U(L) = U(\sum_n p_n L^n) = \sum_n p_n U(L^n) = \sum_n p_n u_n$. Thus, $U(\cdot)$ has the expected utility form.

In the other direction, suppose that $U(\cdot)$ has the expected utility form, and consider any compound lottery $(L_1, \dots, L_K; \alpha_1, \dots, \alpha_K)$, where $L_k = (p_1^k, \dots, p_N^k)$. Its reduced lottery is $L' = \sum_k \alpha_k L_k$. Hence,

$$U\left(\sum_k \alpha_k L_k\right) = \sum_n u_n \left(\sum_k \alpha_k p_n^k\right) = \sum_k \alpha_k \left(\sum_n u_n p_n^k\right) = \sum_k \alpha_k U(L_k).$$

Thus, property (6.B.1) is satisfied. ■

The expected utility property is a *cardinal* property of utility functions defined on the space of lotteries. In particular, the result in Proposition 6.B.2 shows that the expected utility form is preserved only by increasing *linear* transformations.

Proposition 6.B.2: Suppose that $U: \mathcal{L} \rightarrow \mathbb{R}$ is a v.N–M expected utility function for the preference relation $\succsim$ on $\mathcal{L}$. Then $\tilde{U}: \mathcal{L} \rightarrow \mathbb{R}$ is another v.N–M utility function for $\succsim$ if and only if there are scalars $\beta > 0$ and γ such that $\tilde{U}(L) = \beta U(L) + \gamma$ for every $L \in \mathcal{L}$.

Proof: Begin by choosing two lotteries $\bar{L}$ and $\underline{L}$ with the property that $\bar{L} \succ L \succ \underline{L}$ for all $L \in \mathcal{L}$.⁶ If $\bar{L} \sim \underline{L}$, then every utility function is a constant and the result follows immediately. Therefore, we assume from now on that $\bar{L} \succ \underline{L}$.

6. These best and worst lotteries can be shown to exist. We could, for example, choose a maximizer and a minimizer of the linear, hence continuous, function $U(\cdot)$ on the simplex of probabilities, a compact set.

Note first that if $U(\cdot)$ is a v.N-M expected utility function and $\tilde{U}(L) = \beta U(L) + \gamma$, then

$$\begin{aligned}\tilde{U}\left(\sum_{k=1}^K \alpha_k L_k\right) &= \beta U\left(\sum_{k=1}^K \alpha_k L_k\right) + \gamma \\ &= \beta \left[\sum_{k=1}^K \alpha_k U(L_k) \right] + \gamma \\ &= \sum_{k=1}^K \alpha_k [\beta U(L_k) + \gamma] \\ &= \sum_{k=1}^K \alpha_k \tilde{U}(L_k).\end{aligned}$$

Since $\tilde{U}(\cdot)$ satisfies property (6.B.1), it has the expected utility form.

For the reverse direction, we want to show that if both $\tilde{U}(\cdot)$ and $U(\cdot)$ have the expected utility form, then constants $\beta > 0$ and γ exist such that $\tilde{U}(L) = \beta U(L) + \gamma$ for all $L \in \mathcal{L}$. To do so, consider any lottery $L \in \mathcal{L}$, and define $\lambda_L \in [0, 1]$ by

$$U(L) = \lambda_L U(\bar{L}) + (1 - \lambda_L) U(L).$$

Thus

$$\lambda_L = \frac{U(L) - U(\bar{L})}{U(\bar{L}) - U(L)} \quad (6.B.2)$$

Since $\lambda_L U(\bar{L}) + (1 - \lambda_L) U(L) = U(\lambda_L \bar{L} + (1 - \lambda_L)L)$ and $U(\cdot)$ represents the preferences $\succeq$, it must be that $L \sim \lambda_L \bar{L} + (1 - \lambda_L)L$. But if so, then since $\tilde{U}(\cdot)$ is also linear and represents these same preferences, we have

$$\begin{aligned}\tilde{U}(L) &= \tilde{U}(\lambda_L \bar{L} + (1 - \lambda_L)L) \\ &= \lambda_L \tilde{U}(\bar{L}) + (1 - \lambda_L) \tilde{U}(L) \\ &= \lambda_L (\tilde{U}(\bar{L}) - \tilde{U}(L)) + \tilde{U}(L).\end{aligned}$$

Substituting for λ_L from (6.B.2) and rearranging terms yields the conclusion that $\tilde{U}(L) = \beta U(L) + \gamma$, where

$$\beta = \frac{\tilde{U}(\bar{L}) - \tilde{U}(L)}{U(\bar{L}) - U(L)}$$

and

$$\gamma = \tilde{U}(L) - U(L) \frac{\tilde{U}(\bar{L}) - \tilde{U}(L)}{U(\bar{L}) - U(L)}.$$

This completes the proof ■

A consequence of Proposition 6.B.2 is that for a utility function with the expected utility form, differences of utilities have meaning. For example, if there are four outcomes, the statement "the difference in utility between outcomes 1 and 2 is greater than the difference between outcomes 3 and 4," $u_1 - u_2 > u_3 - u_4$, is equivalent to

$$\frac{1}{2}u_1 + \frac{1}{2}u_4 > \frac{1}{2}u_2 + \frac{1}{2}u_3.$$

Therefore, the statement means that the lottery $L = (\frac{1}{2}, 0, 0, \frac{1}{2})$ is preferred to the lottery $L' = (0, \frac{1}{2}, \frac{1}{2}, 0)$. This ranking of utility differences is preserved by all linear transformations of the v.N-M expected utility function.

Note that if a preference relation $\succeq$ on $\mathcal{L}$ is representable by a utility function $U(\cdot)$ that has the expected utility form, then since a linear utility function is continuous, it follows that $\succeq$ is continuous on $\mathcal{L}$. More importantly, the preference relation $\succeq$ must also satisfy the independence axiom. You are asked to show this in Exercise 6.B.2.

Exercise 6.B.2: Show that if the preference relation $\succeq$ on $\mathcal{L}$ is represented by a utility function $U(\cdot)$ that has the expected utility form, then $\succeq$ satisfies the independence axiom.

The expected utility theorem, the central result of this section, tells us that the converse is also true.

The Expected Utility Theorem

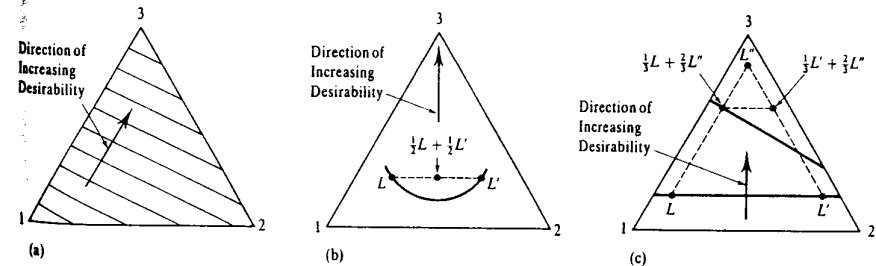
The *expected utility theorem* says that if the decision maker's preferences over lotteries satisfy the continuity and independence axioms, then his preferences are representable by a utility function with the expected utility form. It is the most important result in the theory of choice under uncertainty, and the rest of the book bears witness to its usefulness.

Before stating and proving the result formally, however, it may be helpful to attempt an intuitive understanding of why it is true.

Consider the case where there are only three outcomes. As we have already observed, the continuity axiom insures that preferences on lotteries can be represented by some utility function. Suppose that we represent the indifference map in the simplex, as in Figure 6.B.5. Assume, for simplicity, that we have a conventional map with one-dimensional indifference curves. Because the expected utility form is linear in the probabilities, representability by the expected utility form is equivalent to these indifference curves being straight, parallel lines (you should check this). Figure 6.B.5(a) exhibits an indifference map satisfying these properties. We now argue that these properties are, in fact, consequences of the independence axiom.

Indifference curves are straight lines if, for every pair of lotteries L, L' , we have that $L \sim L'$ implies $\alpha L + (1 - \alpha)L' \sim L$ for all $\alpha \in [0, 1]$. Figure 6.B.5(b) depicts a situation where the indifference curve is not a straight line; we have $L' \sim L$ but

FIGURE 6.B.5 Geometric explanation of the expected utility theorem. (a) $\succeq$ is representable by a utility function with the expected utility form. (b) Contradiction of the independence axiom. (c) Contradiction of the independence axiom.



$\frac{1}{2}L' + \frac{1}{2}L > L$. This is equivalent to saying that

$$\frac{1}{2}L' + \frac{1}{2}L > \frac{1}{2}L + \frac{1}{2}L. \quad (6.B.3)$$

But since $L \sim L'$, the independence axiom implies that we must have $\frac{1}{2}L' + \frac{1}{2}L \sim \frac{1}{2}L + \frac{1}{2}L$ (see Exercise 6.B.1). This contradicts (6.B.3), and so we must conclude that indifference curves are straight lines.

Figure 6.B.5(c) depicts two straight but nonparallel indifference lines. A violation of the independence axiom can be constructed in this case, as indicated in the figure. There we have $L \succ L'$ (in fact, $L \sim L'$), but $\frac{1}{2}L + \frac{1}{2}L' \succ \frac{1}{2}L' + \frac{1}{2}L$ does not hold for the lottery L'' shown in the figure. Thus, indifference curves must be parallel, straight lines if preferences satisfy the independence axiom.

In Proposition 6.B.3, we formally state and prove the expected utility theorem.

Proposition 6.B.3: (Expected Utility Theorem) Suppose that the rational preference relation $\succsim$ on the space of lotteries $\mathcal{L}$ satisfies the continuity and independence axioms. Then $\succsim$ admits a utility representation of the expected utility form. That is, we can assign a number u_n to each outcome $n = 1, \dots, N$ in such a manner that for any two lotteries $L = (p_1, \dots, p_N)$ and $L' = (p'_1, \dots, p'_N)$, we have

$$L \succsim L' \text{ if and only if } \sum_{n=1}^N u_n p_n \geq \sum_{n=1}^N u_n p'_n. \quad (6.B.4)$$

Proof: We organize the proof in a succession of steps. For simplicity, we assume that there are best and worst lotteries in $\mathcal{L}$, $\bar{L}$ and $\underline{L}$ (so, $\bar{L} \succsim L \succsim \underline{L}$ for any $L \in \mathcal{L}$).⁷ If $\bar{L} \sim \underline{L}$, then all lotteries in $\mathcal{L}$ are indifferent and the conclusion of the proposition holds trivially. Hence, from now on, we assume that $\bar{L} > \underline{L}$.

Step 1. If $L > L'$ and $\alpha \in (0, 1)$, then $L > \alpha L + (1 - \alpha)L' > L'$.

This claim makes sense. A nondegenerate mixture of two lotteries will hold a preference position strictly intermediate between the positions of the two lotteries. Formally, the claim follows from the independence axiom. In particular, since $L > L'$, the independence axiom implies that (recall Exercise 6.B.1)

$$L = \alpha L + (1 - \alpha)L > \alpha L + (1 - \alpha)L' > \alpha L' + (1 - \alpha)L' = L'.$$

Step 2. Let $\alpha, \beta \in [0, 1]$. Then $\beta \bar{L} + (1 - \beta)\underline{L} > \alpha \bar{L} + (1 - \alpha)\underline{L}$ if and only if $\beta > \alpha$.

Suppose that $\beta > \alpha$. Note first that we can write

$$\beta \bar{L} + (1 - \beta)\underline{L} = \gamma \bar{L} + (1 - \gamma)[\alpha \bar{L} + (1 - \alpha)\underline{L}],$$

where $\gamma = [(\beta - \alpha)/(1 - \alpha)] \in (0, 1]$. By Step 1, we know that $\bar{L} > \alpha \bar{L} + (1 - \alpha)\underline{L}$. Applying Step 1 again, this implies that $\gamma \bar{L} + (1 - \gamma)[\alpha \bar{L} + (1 - \alpha)\underline{L}] > \alpha \bar{L} + (1 - \alpha)\underline{L}$, and so we conclude that $\beta \bar{L} + (1 - \beta)\underline{L} > \alpha \bar{L} + (1 - \alpha)\underline{L}$.

For the converse, suppose that $\beta \leq \alpha$. If $\beta = \alpha$, we must have $\beta \bar{L} + (1 - \beta)\underline{L} \sim \alpha \bar{L} + (1 - \alpha)\underline{L}$. So suppose that $\beta < \alpha$. By the argument proved in the previous

7. In fact, with our assumption of a finite set of outcomes, this can be established as a consequence of the independence axiom (see Exercise 6.B.3).

paragraph (reversing the roles of α and β), we must then have $\alpha \bar{L} + (1 - \alpha)\underline{L} > \beta \bar{L} + (1 - \beta)\underline{L}$.

Step 3. For any $L \in \mathcal{L}$, there is a unique α_L such that $[\alpha_L \bar{L} + (1 - \alpha_L)\underline{L}] \sim L$.

Existence of such an α_L is implied by the continuity of $\succsim$ and the fact that $\bar{L}$ and $\underline{L}$ are, respectively, the best and the worst lottery. Uniqueness follows from the result of Step 2.

The existence of α_L is established in a manner similar to that used in the proof of Proposition 3.C.1. Specifically, define the sets

$$\{\alpha \in [0, 1] : \alpha \bar{L} + (1 - \alpha)\underline{L} \succeq L\} \quad \text{and} \quad \{\alpha \in [0, 1] : L \succeq \alpha \bar{L} + (1 - \alpha)\underline{L}\}.$$

By the continuity and completeness of $\succsim$, both sets are closed, and any $\alpha \in [0, 1]$ belongs to at least one of the two sets. Since both sets are nonempty and $[0, 1]$ is connected, it follows that there is some α belonging to both. This establishes the existence of an α_L such that $\alpha_L \bar{L} + (1 - \alpha_L)\underline{L} \sim L$.

Step 4. The function $U: \mathcal{L} \rightarrow \mathbb{R}$ that assigns $U(L) = \alpha_L$ for all $L \in \mathcal{L}$ represents the preference relation $\succsim$.

Observe that, by Step 3, for any two lotteries $L, L' \in \mathcal{L}$, we have

$$L \succeq L' \text{ if and only if } \alpha_L \bar{L} + (1 - \alpha_L)\underline{L} \succeq \alpha_{L'} \bar{L} + (1 - \alpha_{L'})\underline{L}.$$

Thus, by Step 2, $L \succeq L'$ if and only if $\alpha_L \geq \alpha_{L'}$.

Step 5. The utility function $U(\cdot)$ that assigns $U(L) = \alpha_L$ for all $L \in \mathcal{L}$ is linear and therefore has the expected utility form.

We want to show that for any $L, L' \in \mathcal{L}$, and $\beta \in [0, 1]$, we have $U(\beta L + (1 - \beta)L') = \beta U(L) + (1 - \beta)U(L')$. By definition, we have

$$L \sim U(L)\bar{L} + (1 - U(L))\underline{L}$$

and

$$L' \sim U(L')\bar{L} + (1 - U(L'))\underline{L}.$$

Therefore, by the independence axiom (applied twice),

$$\beta L + (1 - \beta)L' \sim \beta[U(L)\bar{L} + (1 - U(L))\underline{L}] + (1 - \beta)L'$$

$$\sim \beta[U(L)\bar{L} + (1 - U(L))\underline{L}] + (1 - \beta)[U(L')\bar{L} + (1 - U(L'))\underline{L}].$$

Rearranging terms, we see that the last lottery is algebraically identical to the lottery

$$[\beta U(L) + (1 - \beta)U(L')]\bar{L} + [1 - \beta U(L) - (1 - \beta)U(L')]\underline{L}.$$

In other words, the compound lottery that gives lottery $[U(L)\bar{L} + (1 - U(L))\underline{L}]$ with probability β and lottery $[U(L')\bar{L} + (1 - U(L'))\underline{L}]$ with probability $(1 - \beta)$ has the same reduced lottery as the compound lottery that gives lottery $\bar{L}$ with probability $\beta U(L) + (1 - \beta)U(L')$ and lottery $\underline{L}$ with probability $[1 - \beta U(L) - (1 - \beta)U(L')]$. Thus

$$\beta L + (1 - \beta)L' \sim [\beta U(L) + (1 - \beta)U(L')]\bar{L} + [1 - \beta U(L) - (1 - \beta)U(L')]\underline{L}.$$

By the construction of $U(\cdot)$ in Step 4, we therefore have

$$U(\beta L + (1 - \beta)L') = \beta U(L) + (1 - \beta)U(L'),$$

as we wanted.

Together, Steps 1 to 5 establish the existence of a utility function representing $\succeq$ that has the expected utility form. ■

Discussion of the Theory of Expected Utility

A first advantage of the expected utility theorem is technical: It is extremely convenient analytically. This, more than anything else, probably accounts for its pervasive use in economics. It is very easy to work with expected utility and very difficult to do without it. As we have already noted, the rest of the book attests to the importance of the result. Later in this chapter, we will explore some of the analytical uses of expected utility.

A second advantage of the theorem is normative: Expected utility may provide a valuable guide to action. People often find it hard to think systematically about risky alternatives. But if an individual believes that his choices should satisfy the axioms on which the theorem is based (notably, the independence axiom), then the theorem can be used as a guide in his decision process. This point is illustrated in Example 6.B.1.

Example 6.B.1: Expected Utility as a Guide to Introspection. A decision maker may not be able to assess his preference ordering between the lotteries L and L' depicted in Figure 6.B.6. The lotteries are too close together, and the differences in the probabilities involved are too small to be understood. Yet, if the decision maker believes that his preferences should satisfy the assumptions of the expected utility theorem, then he may consider L'' instead, which is on the straight line spanned by L and L' but at a significant distance from L . The lottery L'' may not be a feasible choice, but if he determines that $L'' > L$, then he can conclude that $L' > L$. Indeed, if $L'' > L$, then there is an indifference curve separating these two lotteries, as shown in the figure, and it follows from the fact that indifference curves are a family of parallel straight lines that there is also an indifference curve separating L' and L , so that $L' > L$. Note that this type of inference is not possible using only the general

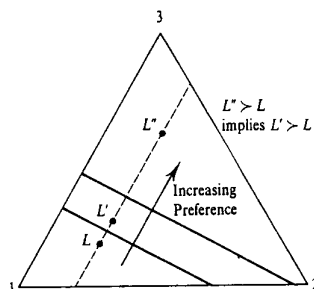


Figure 6.B.6
Expected utility as a
guide to introspection

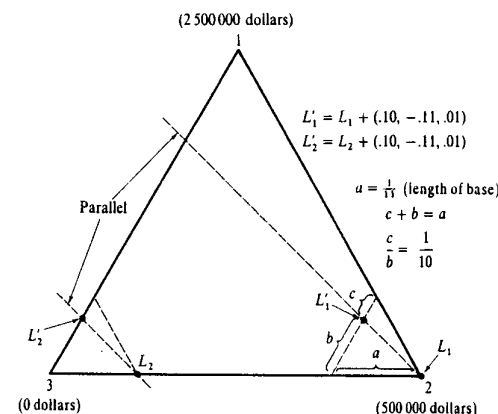


Figure 6.B.7
Depiction of the Allais
paradox in the simplex.

choice theory of Chapter 1 because, without the hypotheses of the expected utility theorem, the indifference curves need not be straight lines (with a general indifference map, we could perfectly well have $L'' > L$ and $L > L'$).

A concrete example of this use of the expected utility theorem is developed in Exercise 6.B.4. ■

As a descriptive theory, however, the expected utility theorem (and, by implication, its central assumption, the independence axiom), is not without difficulties. Examples 6.B.2 and 6.B.3 are designed to test its plausibility.

Example 6.B.2: The Allais Paradox. This example, known as the Allais paradox [from Allais (1953)], constitutes the oldest and most famous challenge to the expected utility theorem. It is a thought experiment. There are three possible monetary prizes (so the number of outcomes is $N = 3$):

First Prize	Second Prize	Third Prize
2 500 000 dollars	500 000 dollars	0 dollars

The decision maker is subjected to two choice tests. The first consists of a choice between the lotteries L_1 and L'_1 :

$$L_1 = (0, 1, 0) \quad L'_1 = (.10, .89, .01).$$

The second consists of a choice between the lotteries L_2 and L'_2 :

$$L_2 = (0, .11, .89) \quad L'_2 = (.10, 0, .90).$$

The four lotteries involved are represented in the simplex diagram of Figure 6.B.7.

It is common for individuals to express the preferences $L_1 > L'_1$ and $L'_2 > L_2$.⁸

8. In our classroom experience, roughly half the students choose this way.

The first choice means that one prefers the certainty of receiving 500 000 dollars over a lottery offering a 1/10 probability of getting five times more but bringing with it a tiny risk of getting nothing. The second choice means that, all things considered, a 1/10 probability of getting 2 500 000 dollars is preferred to getting only 500 000 dollars with the slightly better odds of 11/100.

However, these choices are not consistent with expected utility. This can be seen in Figure 6.B.7: The straight lines connecting L_1 to L'_1 and L_2 to L'_2 are parallel. Therefore, if an individual has a linear indifference curve that lies in such a way that L_1 is preferred to L'_1 , then a parallel linear indifference curve must make L_2 preferred to L'_2 , and vice versa. Hence, choosing L_1 and L'_2 is inconsistent with preferences satisfying the assumptions of the expected utility theorem.

More formally, suppose that there was a v.N-M expected utility function. Denote by u_{25} , u_{05} , and u_0 the utility values of the three outcomes. Then the choice $L_1 > L'_1$ implies

$$u_{05} > (.10)u_{25} + (.89)u_{05} + (.01)u_0.$$

Adding $(.89)u_0 - (.89)u_{05}$ to both sides, we get

$$(.11)u_{05} + (.89)u_0 > (.10)u_{25} + (.90)u_0,$$

and therefore any individual with a v.N-M utility function must have $L_2 > L'_2$. ■

There are four common reactions to the Allais paradox. The first, propounded by J. Marschack and L. Savage, goes back to the normative interpretation of the theory. It argues that choosing under uncertainty is a reflective activity in which one should be ready to correct mistakes if they are proven inconsistent with the basic principles of choice embodied in the independence axiom (much as one corrects arithmetic mistakes).

The second reaction maintains that the Allais paradox is of limited significance for economics as a whole because it involves payoffs that are out of the ordinary and probabilities close to 0 and 1.

A third reaction seeks to accommodate the paradox with a theory that defines preferences over somewhat larger and more complex objects than simply the ultimate lottery over outcomes. For example, the decision maker may value not only what he receives but also what he receives compared with what he might have received by choosing differently. This leads to *regret theory*. In the example, we could have $L_1 > L'_1$ because the expected regret caused by the possibility of getting zero in lottery L'_1 , when choosing L_1 would have assured 500 000 dollars, is too great. On the other hand, with the choice between L_2 and L'_2 , no such clear-cut regret potential exists; the decision maker was very likely to get nothing anyway.

The fourth reaction is to stick with the original choice domain of lotteries but to give up the independence axiom in favor of something weaker. Exercise 6.B.5 develops this point further.

Example 6.B.3: Machina's paradox. Consider the following three outcomes: "a trip to Venice," "watching an excellent movie about Venice," and "staying home." Suppose that you prefer the first to the second and the second to the third.

Now you are given the opportunity to choose between two lotteries. The first lottery gives "a trip to Venice" with probability 99.9% and "watching an excellent movie about Venice" with probability 0.1%. The second lottery gives "a trip to

Venice," again with probability 99.9% and "staying home" with probability 0.1%. The independence axiom forces you to prefer the first lottery to the second. Yet, it would be understandable if you did otherwise. Choosing the second lottery is the rational thing to do if you anticipate that in the event of not getting the trip to Venice, your tastes over the other two outcomes will change: You will be severely *disappointed* and will feel miserable watching a movie about Venice.

The idea of disappointment has parallels with the idea of regret that we discussed in connection with the Allais paradox, but it is not quite the same. Both ideas refer to the influence of "what might have been" on the level of well-being experienced, and it is because of this that they are in conflict with the independence axiom. But disappointment is more directly concerned with what might have been if another outcome of a given lottery had come up, whereas regret should be thought of as regret over a choice not made. ■

Because of the phenomena illustrated in the previous two examples, the search for a useful theory of choice under uncertainty that does not rely on the independence axiom has been an active area of research [see Machina (1987) and also Hey and Orme (1994)]. Nevertheless, the use of the expected utility theorem is pervasive in economics.

An argument sometimes made against the practical significance of violations of the independence axiom is that individuals with such preferences would be weeded out of the marketplace because they would be open to the acceptance of so-called "Dutch books," that is, deals leading to a sure loss of money. Suppose, for example, that there are three lotteries such that $L > L'$ and $L > L''$ but, in violation of the independence axiom, $\alpha L' + (1 - \alpha)L'' > L$ for some $\alpha \in (0, 1)$. Then, when the decision maker is in the initial position of owning the right to lottery L , he would be willing to pay a small fee to trade L for a compound lottery yielding lottery L' with probability α and lottery L'' with probability $(1 - \alpha)$. But as soon as the first stage of this lottery is over, giving him either L' or L'' we could get him to pay a fee to trade this lottery for L . Hence, at that point, he would have paid the two fees but would otherwise be back to his original position.

This may well be a good argument for convexity of the not-better-than sets of $\succeq$, that is, for it to be the case that $L \succeq \alpha L' + (1 - \alpha)L''$ whenever $L \succeq L'$ and $L \succeq L''$. This property is implied by the independence axiom but is weaker than it. Dutch book arguments for the full independence axiom are possible, but they are more contrived [see Green (1987)].

Finally, one must use some caution in applying the expected utility theorem because in many practical situations the final outcomes of uncertainty are influenced by actions taken by individuals. Often, these actions should be explicitly modeled but are not. Example 6.B.4 illustrates the difficulty involved.

Example 6.B.4: Induced preferences. You are invited to a dinner where you may be offered fish (F) or meat (M). You would like to do the proper thing by showing up with white wine if F is served and red wine if M is served. The action of buying the wine must be taken *before* the uncertainty is resolved.

Suppose now that the cost of the bottle of red or white wine is the same and that you are also indifferent between F and M. If you think of the possible outcomes as F and M, then you are apparently indifferent between the lottery that gives F with certainty and the lottery that gives M with certainty. The independence axiom would

then seem to require that you also be indifferent to a lottery that gives F or M with probability $\frac{1}{2}$ each. But you would clearly not be indifferent, since knowing that either F or M will be served with certainty allows you to buy the right wine, whereas, if you are not certain, you will either have to buy both wines or else bring the wrong wine with probability $\frac{1}{2}$.

Yet this example does not contradict the independence axiom. To appeal to the axiom, the decision framework must be set up so that the satisfaction derived from an outcome does not depend on any action taken by the decision maker before the uncertainty is resolved. Thus, preferences should not be induced or derived from *ex ante* actions.⁹ Here, the action “acquisition of a bottle of wine” is taken before the uncertainty about the meal is resolved.

To put this situation into the framework required, we must include the *ex ante* action as part of the description of outcomes. For example, here there would be four outcomes: “bringing red wine when served M,” “bringing white wine when served M,” “bringing red wine when served F,” and “bringing white wine when served F.” For any underlying uncertainty about what will be served, you induce a lottery over these outcomes by your choice of action. In this setup, it is quite plausible to be indifferent among “having meat and bringing red wine,” “having fish and bringing white wine,” or any lottery between these two outcomes, as the independence axiom requires. ■

Although it is not a contradiction to the postulates of expected utility theory, and therefore it is not a serious conceptual difficulty, the induced preferences example nonetheless raises a practical difficulty in the use of the theory. The example illustrates the fact that, in applications, many economic situations do not fit the pure framework of expected utility theory. Preferences are almost always, to some extent, induced.¹⁰

The expected utility theorem does impose some structure on induced preferences. For example, suppose the complete set of outcomes is $B \times A$, where $B = \{b_1, \dots, b_N\}$ is the set of possible realizations of an exogenous randomness and A is the decision maker's set of possible (*ex ante*) actions. Under the conditions of the expected utility theorem, for every $a \in A$ and $b_n \in B$, we can assign some utility value $u_n(a)$ to the outcome (b_n, a) . Then, for every exogenous lottery $L = (p_1, \dots, p_N)$ on B , we can define a derived utility function by maximizing expected utility:

$$U(L) = \max_{a \in A} \sum_n p_n u_n(a).$$

In Exercise 6.B.6, you are asked to show that while $U(L)$, a function on $\mathcal{L}$, need not be linear,

9. Actions taken *ex post* do not create problems. For example, suppose that $u_n(a_n)$ is the utility derived from outcome n when action a_n is taken after the realization of uncertainty. The decision maker therefore chooses a_n to solve $\max_{a_n \in A_n} u_n(a_n)$, where A_n is the set of possible actions when outcome n occurs. We can then let $u_n = \max_{a_n \in A_n} u_n(a_n)$ and evaluate lotteries over the N outcomes as in expected utility theory.

10. Consider, for example, preferences for lotteries over amounts of money available tomorrow. Unless the individual's preferences over consumption today and tomorrow are additively separable, his decision of how much to consume today—a decision that must be made before the resolution of the uncertainty concerning tomorrow's wealth—affects his preferences over these lotteries in a manner that conflicts with the fulfillment of the independence axiom.

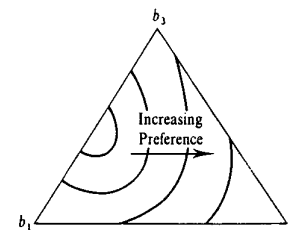


Figure 6.B.8
An indifference map
for induced preferences
over lotteries on
 $B = \{b_1, b_2, b_3\}$.

it is nonetheless always convex; that is,

$$U(\alpha L + (1 - \alpha)L') \leq \alpha U(L) + (1 - \alpha)U(L').$$

Figure 6.B.8 represents an indifference map for induced preferences in the probability simplex for a case where $N = 3$.

6.C Money Lotteries and Risk Aversion

In many economic settings, individuals seem to display aversion to risk. In this section, we formalize the notion of *risk aversion* and study some of its properties.

From this section through the end of the chapter, we concentrate on risky alternatives whose outcomes are amounts of money. It is convenient, however, when dealing with monetary outcomes, to treat money as a continuous variable. Strictly speaking, the derivation of the expected utility representation given in Section 6.B assumed a finite number of outcomes. However, the theory can be extended, with some minor technical complications, to the case of an infinite domain. We begin by briefly discussing this extension.

Lotteries over Monetary Outcomes and the Expected Utility Framework

Suppose that we denote amounts of money by the continuous variable x . We can describe a monetary lottery by means of a *cumulative distribution function* $F: \mathbb{R} \rightarrow [0, 1]$. That is, for any x , $F(x)$ is the probability that the realized payoff is less than or equal to x . Note that if the distribution function of a lottery has a density function $f(\cdot)$ associated with it, then $F(x) = \int_{-\infty}^x f(t) dt$ for all x . The advantage of a formalism based on distribution functions over one based on density functions, however, is that the first is completely general. It does not exclude a priori the possibility of a discrete set of outcomes. For example, the distribution function of a lottery with only three monetary outcomes receiving positive probability is illustrated in Figure 6.C.1.

Note that distribution functions preserve the linear structure of lotteries (as do density functions). For example, the final distribution of money, $F(\cdot)$, induced by a compound lottery $(L_1, \dots, L_K; \alpha_1, \dots, \alpha_K)$ is just the weighted average of the distributions induced by each of the lotteries that constitute it: $F(x) = \sum_1^K \alpha_k F_k(x)$, where $F_k(\cdot)$ is the distribution of the payoff under lottery L_k .

From this point on, we shall work with distribution functions to describe lotteries over monetary outcomes. We therefore take the lottery space $\mathcal{L}$ to be the set of all